Example 7.25

The marks obtained in a certain exam follow normal distribution with mean 45 and SD 10. If 1,300 students appeared at the examination, calculate the number of students scoring (i) less than 35 marks and (ii) more than 65 marks.

Solution :

Let X be the normal variate showing the score of the candidate with mean 45 and standard deviation 10.

(i) less than 35 marks

When $X = 35$

$$Z = \frac{X - \mu}{\sigma} = \frac{35 - 45}{10} = -1$$

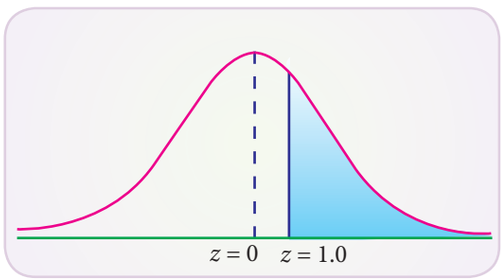


Fig 7.13

$P(X < 35) = P(Z < -1)$

$P(Z > 1) = 0.5 - P(0 < Z < 1)$

$= 0.5 - 0.3413$

$= 0.1587$

Expected number of students scoring less than 35 marks are $0.1587 \times 1300 = 206$

(ii) more than 65 marks

When $X = 65$

$$Z = \frac{X - \mu}{\sigma} = \frac{65 - 45}{10} = 2.0$$

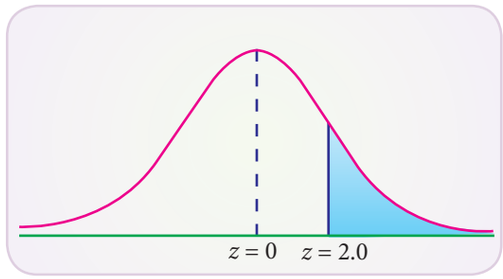


Fig 7.14

$P(X > 65) = P(Z > 2.0)$

$0.5 - P(0 < Z < 2.0)$

$0.5 - 0.4772$

$= 0.0228$

Expected number of students scoring more than 65 marks are 0.0228 x 1300 = 30

Example 7.26

900 light bulbs with a mean life of 125 days are installed in a new factory. Their length of life is normally distributed with a standard deviation of 18 days. What is the expected number of bulbs expire in less than 95 days?

Solution :

Let X be the normal variate of life of light bulbs with mean 125 and standard deviation 18.

(i) less than 95 days

When $X = 95$

$$Z = \frac{X - \mu}{\sigma} = \frac{95 - 125}{18} = -1.667$$

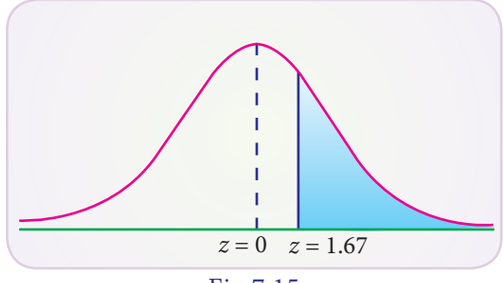


Fig 7.15

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